

CSAI415 — D2 Report: Qdrant Vector Database Integration

Person 2 | Qdrant Migration & Semantic Search

What Was Done

The D1 embeddings were migrated from a local `.npy` file into **Qdrant**, a production-ready vector database hosted on the cloud. This replaces the simple `np.dot` search from D1 with a scalable, queryable vector store that the whole team can connect to.

Setup

- Cluster created on Qdrant Cloud (AWS, Ireland)
- Collection `research_papers` created with 384-dimension vectors and Cosine distance
- All 1,958 chunk embeddings uploaded in batches of 100
- Model used: `all-MiniLM-L6-v2`

Evaluation Results

Component	Collection	Vectors	Vector Size	Distance	Recall@5	P95 Latency	Avg Latency
Qdrant Vector DB	<code>research_papers</code>	1958	384	Cosine	0.70	0.219s	0.230s

Why Recall is Lower Than D1

D1 used exact search (`np.dot`) locally. Qdrant uses **approximate nearest neighbor (HNSW index)** which trades a small amount of accuracy for speed and scalability. This is the standard approach in production retrieval systems and is acceptable for our use case. Recall@5 of 0.70 is above the project's recommended baseline of ≥ 0.60 .

Sample Search Results

Query: *"What is retrieval augmented generation?"*

Rank	Score	Citation
1	0.694	retrieval augmented generation for large language models (p.17)
2	0.662	retrieval augmented multimodal language modeling (p.14)
3	0.638	can llms evaluate retrieval augmented generation (p.8)

Results are semantically accurate and relevant. 

Conclusion

All 1,958 vectors are now live in Qdrant cloud and searchable by meaning. Person 3 can connect directly using the credentials provided in the notebook to build hybrid retrieval on top of this foundation.